

AL HASBY NAZRIL ILHAM PERSONAL PORTFOLIO CMS PRD v1.0 SRS v1.0 TDD v1.0 FINAL SOFTWARE DOCUMENTATION

Document Status : FINAL DOCUMENTATION

Version : 1.0

Project Type : Full Stack Portfolio CMS

Frontend : React.js

Backend : Node.js Express

Database : PostgreSQL

Dokumen ini menjadi blueprint resmi pengembangan website portfolio profesional dengan sistem Content Management System (CMS).

DOCUMENT CONTROL

Version 1.0

Status: Final

Purpose: Menjadi acuan desain, pengembangan, testing, dan deployment aplikasi Portfolio CMS.

1. EXECUTIVE SUMMARY

AL Hasby Nazril Ilham Personal Portfolio CMS merupakan aplikasi portfolio digital modern yang menggabungkan website public dan Admin Dashboard.

Website public digunakan untuk menampilkan profil, pendidikan, skill, project, sertifikat, dan informasi kontak.

Admin Dashboard digunakan untuk mengelola seluruh konten tanpa melakukan perubahan source code.

2. BACKGROUND & PROBLEM STATEMENT

Permasalahan:

- Portfolio statis sulit diperbarui.
- Perubahan data membutuhkan edit kode.
- Tidak ada sistem pengelolaan konten.
- Data project dan sertifikat tidak terstruktur.

Solusi:

Membangun CMS portfolio yang dinamis dan scalable.

3. PRODUCT VISION & GOALS

Vision:

Membangun personal branding platform digital yang profesional untuk bidang teknologi, matematika, dan software development.

Goals:

- Menampilkan portfolio profesional.
- Mendukung kebutuhan internship dan career preparation.
- Memudahkan pengelolaan konten.
- Membuat sistem yang mudah dikembangkan.

4. USER ANALYSIS

Visitor:

- Recruiter
- Developer
- Client
- Akademisi

Tujuan:

Melihat pengalaman, kemampuan, dan hasil karya.

Admin:

Pemilik portfolio yang mengelola semua informasi website.

5. PUBLIC WEBSITE REQUIREMENT

Loading Screen:

- Branding animation
- Transition effect

Home:

- Nama
- Profesi
- Deskripsi
- Statistik
- Social media

- Interactive Lanyard Card

Lanyard:

- Drag interaction
- Physics animation
- Spring effect
- Responsive touch

About:

- Biography
- Profile image

Education:

- Timeline pendidikan

Skills:

- Programming
- Framework
- Tools

Projects:

- Project card
- Category
- Technology stack
- Github link
- Demo link

Certificate:

- Gallery sertifikat

Contact:

- Form komunikasi

6. ADMIN CMS REQUIREMENT

Dashboard:

- Statistik project
- Statistik sertifikat
- Pesan masuk

Profile Management:

- Edit nama
- Edit profesi
- Edit bio
- Upload foto
- Upload CV

Project Management:

- Create project
- Update project
- Delete project
- Upload image
- Category management
- Technology management
- Github URL
- Demo URL

Education:

CRUD data pendidikan

Skill:

CRUD skill dan persentase

Certificate:

CRUD sertifikat

Social Media:

- Instagram
- Github
- LinkedIn
- WhatsApp

7. FUNCTIONAL REQUIREMENT ID

HOME-001: Menampilkan halaman home.

HOME-002: Menampilkan lanyard interactive card.

PROFILE-001: Admin mengubah profil.

PROJECT-001: Admin CRUD project.

EDU-001: Admin CRUD pendidikan.

SKILL-001: Admin CRUD skill.

CERT-001: Admin CRUD sertifikat.

CONTACT-001: Visitor mengirim pesan.

AUTH-001: Admin login menggunakan authentication.

8. AUTHENTICATION DESIGN

Flow:

Admin membuka login.

Input email dan password.

Backend melakukan validasi.

JWT token dibuat.

Admin masuk dashboard.

Security:

- Password hashing
- JWT authentication
- Session management
- Permission middleware.

9. DATABASE DESIGN

Users:

id, email, password, role

Profile:

id, name, profession, bio, image, cv_url

Projects:

id, title, image, category, description, github_url, demo_url, featured

Project Technology:

id, project_id, technology_name

Education:

id, institution, start_date, end_date

Skills:

id, name, category, percentage

Certificate:

id, title, issuer, image, credential_url

Messages:

id, name, email, message, status

10. API SPECIFICATION

Authentication:

POST /api/auth/login

POST /api/auth/logout

Profile:

GET /api/profile

PUT /api/profile

Projects:

GET /api/projects

POST /api/projects

PUT /api/projects/:id

DELETE /api/projects/:id

Education:

GET /api/education

POST /api/education

Skills:

GET /api/skills

POST /api/skills

Certificate:

GET /api/certificates

POST /api/certificates

Messages:

GET /api/messages

11. FRONTEND ARCHITECTURE

React Structure:

src

- assets
- components
- pages
- layouts
- services
- hooks
- context
- utils
- routes

Main Components:

- Navbar
- Footer
- LanyardCard
- ProjectCard
- Timeline
- SkillCard
- AdminSidebar
- DataTable

12. UI DESIGN SYSTEM

Design:

- Dark modern theme
- Glassmorphism
- Smooth animation
- Responsive layout

Components:

- Button
- Card
- Modal
- Form
- Table
- Sidebar

13. SECURITY REQUIREMENT

Security implementation:

- JWT
- Password encryption
- Input validation
- File upload validation
- CORS protection
- Rate limiting
- XSS prevention
- SQL injection prevention

14. TESTING DOCUMENT

Testing:

- Unit Testing
- Integration Testing
- API Testing
- Responsive Testing
- User Acceptance Testing

15. DEPLOYMENT ARCHITECTURE

Architecture:

User

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React Frontend (Vercel)

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Backend API

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PostgreSQL Database

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Cloud Storage

Deployment:

Frontend: Vercel

Backend: Railway/Render

Database: Supabase

Storage: Cloudinary

16. DEVELOPMENT ROADMAP

Sprint 1:

Project setup dan design system

Sprint 2:

Public website

Sprint 3:

Backend API

Sprint 4:

Admin CMS

Sprint 5:

Testing

Sprint 6:

Deployment

17. FUTURE DEVELOPMENT

Future features:

- Blog CMS
- Visitor Analytics
- Multi Admin
- AI Portfolio Assistant
- Theme Customization

18. FINAL CHECKLIST

Completed Requirement:

- ✓ React.js Portfolio
- ✓ Interactive Lanyard
- ✓ Admin CMS

- ✓ Dynamic Content
- ✓ Project Management
- ✓ Certificate Management
- ✓ Education Management
- ✓ Authentication
- ✓ Database Design
- ✓ API Planning
- ✓ Security Planning
- ✓ Testing Planning
- ✓ Deployment Planning